```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model
model = Sequential([
    Dense(10, activation='softmax', input_shape=(784,))
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

model.fit(X_train, y_train, epochs=10, batch_size=64)

test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input_sha
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential"
```

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 10)	7,850

```
 Total params: 7,850 (30.66 KB)
 Trainable params: 7,850 (30.66 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.7757 - loss: 0.8700
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9073 - loss: 0.3380
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9143 - loss: 0.3059
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9197 - loss: 0.2841
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9237 - loss: 0.2725
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9260 - loss: 0.2661
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9275 - loss: 0.2603
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9274 - loss: 0.2635
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9287 - loss: 0.2566
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9291 - loss: 0.2490
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - accuracy: 0.9148 - loss: 0.3042
Test accuracy: 0.9251999855041504
```

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import numpy as np
```

```
# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model
model = Sequential([
    Dense(10, activation='softmax', input_shape=(784,))
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

model.fit(X_train, y_train, epochs=16, batch_size=64)

test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)
```

Model: "sequential_2"

Layer (type)	Output Shape	Param #
dense_2 (Dense)	(None, 10)	7,850

Total params: 7,850 (30.66 KB)
Trainable params: 7,850 (30.66 KB)
Non-trainable params: 0 (0.00 B)

```
Epoch 1/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.7777 - loss: 0.8620
Epoch 2/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9091 - loss: 0.3306
Epoch 3/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9169 - loss: 0.3026
Epoch 4/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9191 - loss: 0.2898
Epoch 5/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9228 - loss: 0.2750
Epoch 6/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9229 - loss: 0.2737
Epoch 7/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9263 - loss: 0.2669
Epoch 8/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9280 - loss: 0.2602
Epoch 9/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9301 - loss: 0.2536
Epoch 10/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9311 - loss: 0.2518
Epoch 11/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9299 - loss: 0.2514
Epoch 12/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9296 - loss: 0.2531
Epoch 13/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9313 - loss: 0.2487
Epoch 14/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9306 - loss: 0.2517
Epoch 15/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9320 - loss: 0.2474
Epoch 16/16
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9335 - loss: 0.2404
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.9161 - loss: 0.3006
Test accuracy: 0.9258000254631042
```

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()
```

```
# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model
model = Sequential([
    Dense(10, activation='softmax', input_shape=(784,))
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

model.fit(X_train, y_train, epochs=10, batch_size=28)

test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)
```

Model: "sequential_3"

Layer (type)	Output Shape	Param #
dense_3 (Dense)	(None, 10)	7,850

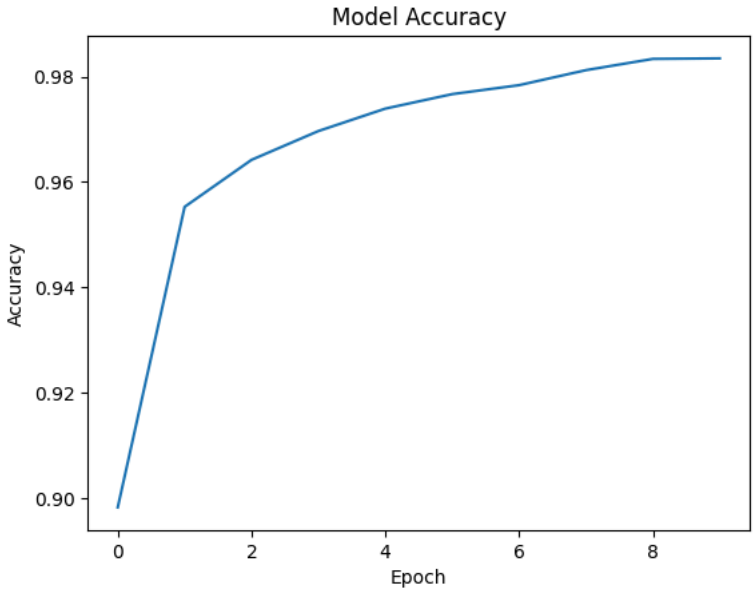
Total params: 7,850 (30.66 KB)
Trainable params: 7,850 (30.66 KB)
Non-trainable params: 0 (0.00 B)

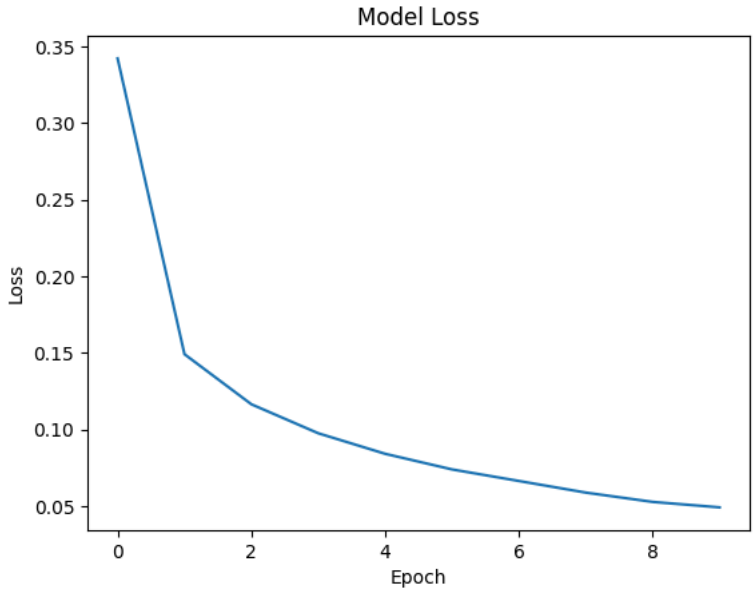
```
Epoch 1/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 5s 2ms/step - accuracy: 0.8166 - loss: 0.7038
Epoch 2/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9158 - loss: 0.3042
Epoch 3/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9192 - loss: 0.2882
Epoch 4/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 5s 2ms/step - accuracy: 0.9224 - loss: 0.2743
Epoch 5/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9264 - loss: 0.2611
Epoch 6/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9295 - loss: 0.2561
Epoch 7/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9295 - loss: 0.2569
Epoch 8/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 6s 3ms/step - accuracy: 0.9308 - loss: 0.2548
Epoch 9/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 4s 2ms/step - accuracy: 0.9302 - loss: 0.2478
Epoch 10/10
2143/2143 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.9306 - loss: 0.2501
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 1ms/step - accuracy: 0.9160 - loss: 0.2988
Test accuracy: 0.9265000224113464
```

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model with 3 hidden layers
```

```
model = Sequential([
    Dense(32, activation='relu', input_shape=(784,)),
    Dense(64, activation='relu'),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train model
history = model.fit(X_train, y_train, epochs=10, batch_size=64)

# Evaluate
test_loss, test_acc = model.evaluate(X_test, y_test)

print("Test Accuracy:", test_acc)

# Plot Accuracy Graph
plt.figure()
plt.plot(history.history['accuracy'])
plt.title('Model Accuracy')
plt.xlabel('Epoch')
plt.ylabel('Accuracy')
plt.show()

# Plot Loss Graph
plt.figure()
plt.plot(history.history['loss'])
plt.title('Model Loss')
plt.xlabel('Epoch')
plt.ylabel('Loss')
plt.show()
```

```
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 6s 4ms/step - accuracy: 0.8105 - loss: 0.6226
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9515 - loss: 0.1617
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9627 - loss: 0.1218
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9704 - loss: 0.0956
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9742 - loss: 0.0858
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 3ms/step - accuracy: 0.9775 - loss: 0.0708
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9806 - loss: 0.0604
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 3ms/step - accuracy: 0.9827 - loss: 0.0560
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9845 - loss: 0.0495
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 4s 5ms/step - accuracy: 0.9853 - loss: 0.0447
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.9649 - loss: 0.1247
Test Accuracy: 0.970300018787384
```






```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, Flatten
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
```

```
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)

# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model
model = Sequential([
    Dense(10, activation='softmax', input_shape=(784,)),
    Dense(10, activation='relu', input_shape=(128,)),
    Dense(10, activation='relu', input_shape=(64,))
])

model.compile(optimizer='adam',
              loss='categorical_crossentropy',
              metrics=['accuracy'])

model.summary()

model.fit(X_train, y_train, epochs=10, batch_size=128)

test_loss, test_acc = model.evaluate(X_test, y_test)
print("Test accuracy:", test_acc)
```

```
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:93: UserWarning: Do not pass an `input_sha
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Model: "sequential_6"
```

Layer (type)	Output Shape	Param #
dense_8 (Dense)	(None, 10)	7,850
dense_9 (Dense)	(None, 10)	110
dense_10 (Dense)	(None, 10)	110

```
 Total params: 8,070 (31.52 KB)
 Trainable params: 8,070 (31.52 KB)
 Non-trainable params: 0 (0.00 B)
Epoch 1/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 3s 4ms/step - accuracy: 0.3275 - loss: 5.8505
Epoch 2/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.5869 - loss: nan
Epoch 3/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.0986 - loss: nan
Epoch 4/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.1017 - loss: nan
Epoch 5/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.0975 - loss: nan
Epoch 6/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.0977 - loss: nan
Epoch 7/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.0985 - loss: nan
Epoch 8/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.0992 - loss: nan
Epoch 9/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 1s 3ms/step - accuracy: 0.0979 - loss: nan
Epoch 10/10
469/469 ━━━━━━━━━━━━━━━━━━━━ 2s 4ms/step - accuracy: 0.0973 - loss: nan
313/313 ━━━━━━━━━━━━━━━━━━━━ 1s 2ms/step - accuracy: 0.0924 - loss: nan
Test accuracy: 0.09799999743700027
```

```
import tensorflow as tf
from tensorflow.keras.datasets import mnist
from tensorflow.keras.utils import to_categorical
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense
import matplotlib.pyplot as plt
import numpy as np

# Load dataset
(X_train, y_train), (X_test, y_test) = mnist.load_data()

# Normalize
X_train = X_train / 255.0
X_test = X_test / 255.0

# Flatten
X_train = X_train.reshape(-1, 784)
X_test = X_test.reshape(-1, 784)
```

```
# One-hot encoding
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# Model with 3 hidden layers
model = Sequential([
    Dense(32, activation='relu', input_shape=(784,)),
    Dense(64, activation='relu'),
    Dense(128, activation='relu'),
    Dense(10, activation='softmax')
])

model.compile(
    optimizer='adam',
    loss='categorical_crossentropy',
    metrics=['accuracy']
)

# Train model
history = model.fit(X_train, y_train, epochs=10, batch_size=64)

# predicting an image

# pick any index
index = 5

# get image
img = X_test[index].reshape(28, 28)

# show image
plt.imshow(img, cmap='gray')
plt.title("Actual Label: " + str(np.argmax(y_test[index])))
plt.show()

# prediction
prediction = model.predict(X_test[index].reshape(1, 784))

print("Predicted Digit:", np.argmax(prediction))
```

```
Downloading data from https://storage.googleapis.com/tensorflow/tf-keras-datasets/mnist.npz
11490434/11490434 ━━━━━━━━━━━━━━━━━━━━ 0s 0us/step
/usr/local/lib/python3.12/dist-packages/keras/src/layers/core/dense.py:106: UserWarning: Do not pass an `input_sh
  super().__init__(activity_regularizer=activity_regularizer, **kwargs)
Epoch 1/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 2ms/step - accuracy: 0.8945 - loss: 0.3546
Epoch 2/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9540 - loss: 0.1571
Epoch 3/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9628 - loss: 0.1216
Epoch 4/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 3s 3ms/step - accuracy: 0.9694 - loss: 0.1001
Epoch 5/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9733 - loss: 0.0874
Epoch 6/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9765 - loss: 0.0744
Epoch 7/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9780 - loss: 0.0666
Epoch 8/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9806 - loss: 0.0603
Epoch 9/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9827 - loss: 0.0536
Epoch 10/10
938/938 ━━━━━━━━━━━━━━━━━━━━ 2s 2ms/step - accuracy: 0.9854 - loss: 0.0458
```

Actual Label: 1